

Weekly Diary – Week 12

Internship Organization: Vikaspedia

Duration: 08/04/2026 – 13/04/2026

Objective of the Week

The primary objective for this week was to design a structured system architecture for capturing, storing, and processing guest user fingerprints along with their associated event data. The focus was on defining how data would flow from the client side to the backend and how it would be efficiently stored and managed for analysis and tracking purposes.

Date – 08 April

On this day, the overall workflow of the fingerprint tracking system was designed. A clear plan was developed to outline how the fingerprint ID would be generated on the client side and then transmitted securely to the backend server. The complete data flow, including user interaction capture and communication between components, was carefully structured to ensure efficiency and reliability.

Date – 09 April

The database schema was designed for storing guest session information using MySQL. Key fields such as `fingerprint_id`, `first_seen`, `last_seen`, and event-related data were defined. The structure was planned to support efficient querying and tracking of user activities over multiple sessions while maintaining proper organization of data.

Date – 10 April

Different types of user events were identified and defined, including page visits, user clicks, and navigation behavior within the application. A structured format for storing event data was created to ensure consistency, making it easier to store, retrieve, and analyze user interactions effectively.

Date – 11 April

API endpoints were designed to handle the transmission of fingerprint and event data from the frontend to the backend system. The APIs were structured to ensure scalability, allowing the system to handle multiple users efficiently while maintaining smooth data processing and minimal delays.

Date – 12 April

The FingerprintJS library was integrated into the frontend application. This enabled the system to generate a unique fingerprint ID for each user session based on browser and device characteristics. Initial tests confirmed that the integration was successful and functioning as expected.

Date – 13 April

Testing was conducted across different browsers and devices to evaluate the consistency of fingerprint generation. The results were analyzed to identify any variations and ensure that the generated identifiers remained stable under different conditions. This step helped in validating the reliability of the fingerprinting mechanism.